

SentinelWire™ SOC Toolkit Handbook v1.0

0) Preparation Phase

- Start capture during suspicious event
- Save PCAP
- Note system/user/environment details

1) MITM MASTER – First Pass Detection

- Detects ARP/DNS/TCP tampering etc.
- 0 packets=Safe, 50+=Likely MITM

2) ARP Spoof Detection

- Router MAC/IP inconsistencies
- Multiple MAC claiming router → MITM

3) DNS Manipulation Check

- Non-standard resolvers
- Long DNS queries
- NXDOMAIN flood

4) TLS/SSL Anomaly Analysis

- No-SNI traffic
- Tiny SYN (40–70 bytes)
- RST floods

5) TCP Tampering Indicators

- Retransmissions
- Out-of-order evidence

6) WLAN Attack Detection

- Deauth attacks
- Probe floods
- Rogue AP signals

7) 15-Second MITM Decision Matrix

★★★ MITM MASTER

★★★ ARP SPOOF

★★★ NON-ROUTER TRAFFIC

8) 30-Second SOC Workflow

1. MITM MASTER
2. ARP
3. DNS

4. TLS

5. TCP

6. Extract IoCs

Threat Actor Mapping (TTP Correlation)

This section maps observable Wireshark anomalies to known threat actors based on TTPs.

- APT28 (Fancy Bear)
 - Prefers DNS tunneling, long DNS queries
 - TLS No-SNI fingerprints
 - Sharp ARP poisoning in lateral movement
- APT29 (Cozy Bear)
 - Low-volume C2 beacons
 - Short SYN packets (40–70 bytes)
 - TLS impersonation with malformed extensions
- Lazarus Group
 - NXDOMAIN-based DGA
 - ARP spoof → forced gateway redirection
 - Out-of-order injection patterns
- FIN7
 - DNS-over-TCP usage
 - TLS anomalies (self-signed C2)
 - Retransmission storms
- Sandworm Team
 - Aggressive ARP spoof waves
 - Beacon bursts every 1–3 seconds
 - Fake router MAC swapping

MITRE ATT&CK; Technique Mapping

This section aligns Wireshark-detectable behaviors with MITRE ATT&CK; techniques.

- T1040 – Network Sniffing

Evidence: Deauth, Probe floods, MITM MASTER hits.

- T1557 – Adversary-in-the-Middle

Evidence: ARP-SPOOF, NON-ROUTER anomalies, dual MACs for gateway.

- T1071.004 – DNS (C2 Channel)

Evidence: LONG DNS, NXDOMAIN, DNS-SUSPECT.

- T1095 – Non-Application Layer Protocol

Evidence: TLS No-SNI, tiny SYN packets (beaconing).

- T1046 – Network Service Scanning

Evidence: SYN sequences, out-of-order scanning bursts.

- T1020 – Data Exfiltration

Evidence: Long DNS → DNS channel exfil.

- T1566 – Phishing → Payload Exec → C2

Evidence: TLS anomalies, C2 retrieval attempts.

SOC v2.0 Detection Playbooks

Standardized SOC playbooks for rapid response.

Playbook 1 – MITM Suspected

1. Run MITM MASTER
2. Check ARP-SPOOF results
3. Validate router MAC manually
4. Run NON-ROUTER traffic filter
5. Extract IoCs
6. Block rogue MAC at router

Playbook 2 – DNS Hijack

1. Run DNS-SUSPECT
2. Identify unknown DNS responders
3. Validate against OS DNS settings
4. Check LONG DNS for C2/DGA
5. Block suspicious IPs
6. Reset network adapter DNS cache

Playbook 3 – TLS Anomaly

1. Run TLS-ANOMALY
2. Check for NO-SNI entries
3. Extract JA3/JA3S fingerprints
4. Compare with malware database
5. Block C2 indicators

Playbook 4 – TCP Tampering

1. Run TCP-RETRANS
2. Run TCP-OOO

3. Validate route consistency
4. Look for ARP anomalies
5. Escalate if combined evidence exists